

SaaS Customer Health & Churn Risk — Executive Summary

Data-driven insights into customer risk profiles, plan types, and regional revenue growth

Total Revenue	Total Customers	Churn Rate	Avg LTV
\$575.065K	1,000	29.30%	\$575.06

Approach

Synthetic SaaS subscription data (1,000 customers, ~1,000 subscriptions, ~18,900 transactions) was generated in SQLite, feature-engineered with SQL window functions (RFM: recency, frequency, monetary value; tenure; regional benchmarking), then modeled in Python. Customers were segmented with K-Means clustering on RFM features, and a Random Forest classifier was trained to predict churn probability from recency, frequency, revenue, tenure, plan price, and RFM score. Results were published to an interactive Power BI dashboard.

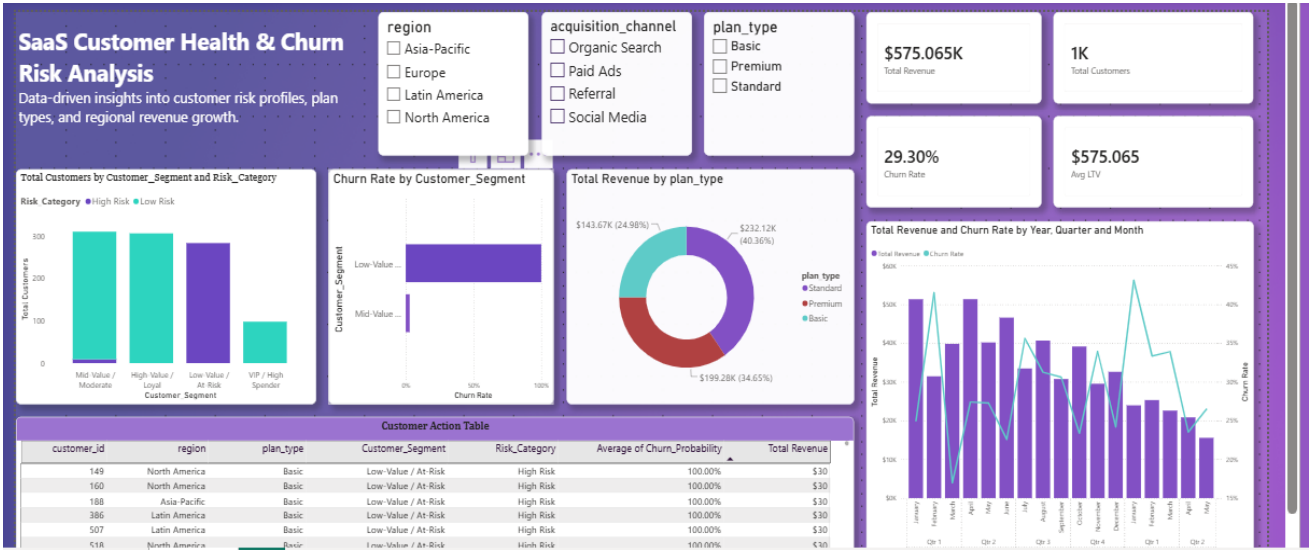
Key Findings

- Overall churn rate is 29.30% across the customer base of 1,000 accounts.
- The Standard plan drives the largest share of revenue (\$232.12K, 40.4%), followed by Premium (\$199.28K, 34.7%) and Basic (\$143.67K, 25.0%).
- The 'Low-Value / At-Risk' customer segment shows the highest churn concentration, with several Basic-plan accounts flagged at ~100% predicted churn probability.
- 'Mid-Value / Moderate' and 'High-Value / Loyal' are the two largest segments by customer count, while 'VIP / High Spender' is the smallest but highest-value tier.
- Revenue and churn rate both show noticeable month-to-month volatility, with churn spikes not always aligned with revenue dips — suggesting churn is concentrated in specific low-spend cohorts rather than broad-based.

Recommended Actions

- Prioritize retention outreach (discounts, plan-downgrade alternatives, proactive support) for Basic-plan customers in the Low-Value/At-Risk segment flagged High Risk by the model.
- Investigate the drivers behind the highest churn-rate months in the trend chart to identify seasonal or campaign-related causes.
- Test upsell paths from Basic to Standard for engaged customers, since Standard carries both higher revenue share and (implicitly) better retention.

Dashboard Preview



Full interactive report: [power_bi/SaaS_Customer_Health.pbix](#) | Source data & code: [data/](#), [sql/](#), [python/](#)